SAULO BARRETO

Senior AI Engineer | LLM Evaluation & Applied Research

sbarreto.cardoso@gmail.com | Brazil (Remote) | linkedin.com/in/saulobarreto | Google Scholar

SUMMARY

Researcher and AI/ML engineer whose trajectory has repeatedly returned to education: teaching assistant at Universidade de Brasília (2013-2014) and, a decade later, Data Science Instructor at Université de Lorraine (2024-2025), alongside 25+ published technical tutorials (Baeldung), and, most recently, an AI Engineer designing the evaluation backbone (LLM-as-judge design, rubric development, benchmark construction) for production LLM systems. Comfortable moving between rigorous research (PhD in high-dimensional signal/image processing, peer-reviewed publications) and shipped software (CI/CD, Docker, tested and documented pipelines). Worked across health, education, mathematics, political, commerce, robotics and finance-adjacent domains, routinely translating technical work for non-technical stakeholders: clients, subject-matter experts, annotator teams, students. Genuinely drawn to Education and human learning, and looking to bring LLM evaluation, applied GenAI, and cross-functional delivery into products that reach students, teachers and families at scale.

CORE SKILLS

- **LLM Evaluation & GenAI:** LLM-as-judge design, eval harness configuration, eval config design, model configuration, Harbor, rubric & benchmark development, evaluation metrics, annotation-platform design, prompt design for structured tasks, model quality auditing
- **Software & Delivery Engineering:** Python (10 yrs), SQL, JavaScript/C++ (working), Git, CI/CD, Docker, containerized reproducible pipelines, automated testing, technical documentation
- **Applied Research & Data Science:** Statistical learning, computer vision (CNNs), tensor decomposition & high-dimensional data analysis, computational optimization, NLP fundamentals
- **Stakeholder Communication & Mentoring:** Cross-functional collaboration with product/business stakeholders and subject-matter experts; led and mentored a 40-person annotator team on quality, process and payments; teaching and instructing experience spanning university courses and technical tutorials (2013-2014, 2021-2025)

EXPERIENCE

Operations Specialist, AI Engineering & LLM Evaluation | Vetto AI

June 2026 - Present

Remote

- **LLM evaluation & judge design:** led research spanning LLM-as-judge design, rubric development, and benchmark construction, running reproducible, isolated model evaluations in Docker-based containerized environments.
- **Production annotation platform:** built and deployed a new annotation platform to production using CI/CD practices to manage releases, mirroring the evaluation/validation pipeline pattern the role's evals and quality workflows require.
- **Data-driven quality workflows:** used SQL to query, structure and validate evaluation and annotation data supporting quality and reporting workflows.
- **Team leadership & mentoring:** led a 40-annotator campaign (computer vision domain), owning quality control, payments and stakeholder communication, translating technical evaluation criteria into clear guidance for a distributed team.
- **Operational tooling:** closed support tickets, built automations, developed dashboards, and organized workflows to streamline day-to-day operations.

Quality Assurance Specialist, LLM Data & Evaluation | Independent Contractor

October 2025 - May 2026

Remote

- **Multi-domain LLM quality:** drove LLM data quality across multiple AI companies, contributing to training, evaluation and benchmarking initiatives spanning education, health, political, mathematics and commerce domains, requiring constant context-switching between stakeholders and domain experts.
- Evaluated 170+ multi-domain datasets against project specifications to ensure data integrity and training quality.
- **Rubric & prompt design:** reviewed and audited 200+ graduate-level mathematics prompts stress-testing state-of-the-art LLMs; led prompt design and rubric development for a commerce-domain evaluation project.
- Designed domain-specific prompts for structured coding tasks; built tests and refactored code to validate outputs; contributed to a model benchmarking initiative using a structured evaluation framework.

Data Science Instructor | Université de Lorraine

February 2024 - November 2025

Nancy, France

- **Education focus:** delivered 84 hours of practical Python and MATLAB sessions to graduate and undergraduate students across NLP, cognitive science, law & economics, and engineering programs, translating advanced technical material for mixed, non-CS audiences.
- Taught data science, machine learning, probability & statistics, information theory, and practical optimization/neural networks.

PhD Student / Researcher | CRAN, Research Centre for Automatic Control

October 2022 - November 2025

Nancy, France

- **Open-source research software:** designed and built pyBBTD, an open-source Python library for high-dimensional imaging data (biomedical imaging, radar) using tensor methods, focused on scalable performance.
- Set up automated testing and CI/CD, learning the toolchain from scratch as part of the project.
- Designed and implemented algorithms integrating linear algebra, optimization and statistical modeling for large-scale data analysis.

Author & Technical Writer | Baeldung

August 2021 - May 2025

Remote

- **Making complex topics accessible:** wrote 25+ tutorials and articles on math for data science, machine learning, and programming, collaborating with five editors, gaining direct, sustained practice explaining technical concepts to a broad, mixed-expertise audience.

Earlier Experience

- **Research Trainee, Nokia Bell Labs** (Budapest, 2022): 3D scene reconstruction for metaverse applications; built a web-based real-time visualization platform (C++, Three.js, Python).
- **Developer, University of Debrecen** (2021): built a pilot vital-signs monitoring system (ECG/EKG) with ML-based high-risk condition detection.
- **Data Annotator, TransPerfect** (2021): labeled data on a 100+ member team for an OCR project under strict ML training guidelines.
- **IT & Communication Coordination, Ministério Público Federal** (Brasília, 2017-2019): produced training documentation for federal prosecutors/lawyers; built an IoT data-center monitoring system (Python, MQTT).
- **Junior Researcher, Antal Bejczy Center for Intelligent Robotics** (Budapest, 2014-2015): humanoid-robot visual pattern recognition (NAO SDK) and a laparoscopic training simulator (OpenCV/PyQt).
- **Teaching Assistant, Universidade de Brasília** (2013-2014): Computer Vision Principles, Algorithms & Data Structures, Experimental Physics.

EDUCATION

- PhD, Automatique, Traitement du Signal et des Images, Génie Informatique, Université de Lorraine, 2025
- M.Sc., Mechatronics, Robotics & Automation Engineering, University of Debrecen, 2020-2022
- B.Sc., Mechatronics, Robotics & Automation Engineering, Universidade de Brasília, 2012-2018

LANGUAGES

Portuguese (Native) | English (Full Professional) | French (Professional Working)